when this light was emitted, the universe was a lot smaller. We illustrate this in Figure 8.7b in which we have put our current *now* slice on the right-hand side of the loaf and included a sequence of slices to the left that depict our universe at ever earlier moments of time. As you can see, the overall size of space and the separations between individual galaxies both decrease as we look at the universe at ever earlier moments.

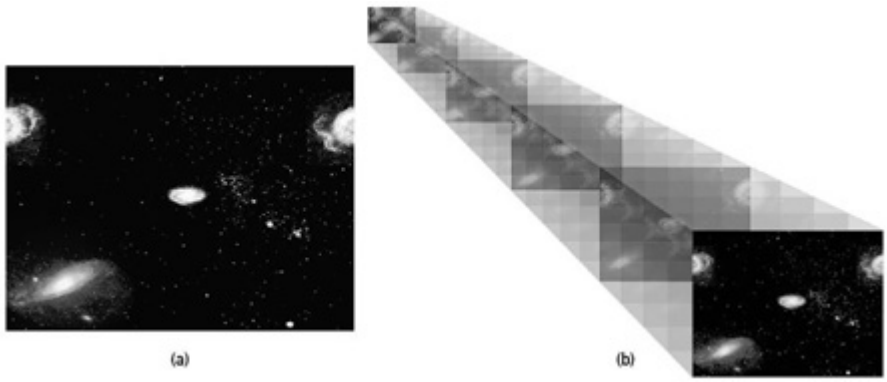


Figure 8.7 **(**a**)** A schematic image depicting all of space right now, assuming space is flat and finite in extent, i.e. shaped like a video game screen. Note that the galaxy on the upper right wraps around on the left. **(**b**)** A schematic image depicting all of space as it evolves through time, with a few time slices emphasized for clarity. Note that the overall size of space and the separation between galaxies decrease as we look farther back in time.

In Figure 8.8, you can also see the history of light, emitted by a distant galaxy perhaps a billion years ago, as it has traveled toward us here in the Milky Way. On the initial slice in Figure 8.8a, the light is first emitted, and on subsequent slices you can see the light getting closer and closer even as the universe gets larger and larger, and finally you can see it reaching us on the rightmost time slice. In Figure 8.8b, by connecting the locations on each slice that the light's leading edge passed through during its journey, we show the light's path through spacetime. Since we receive light from many directions, Figure 8.8c shows a sample of trajectories through space and time that various light beams take to reach us now.